Class Animal

- Variables

```
class Animal {
    7 usages
    private static String food;
    7 usages
    private static int lifeExpectancy;
```

Class Animal

- Constructors

```java
    // Default constructor
    4 usages
    public Animal() {
        this.food = "unknown";
        this.lifeExpectancy = 0;
    }

    // Constructor for food
    1 usage
    public Animal(String food) {
        this.food = food;
        this.lifeExpectancy = 0;
    }

    // Constructor for life expectancy
    1 usage
    public Animal(int lifeExpectancy) {
        this.food = "unknown";
        this.lifeExpectancy = lifeExpectancy;
    }

    // Constructor for both food and life expectancy
    1 usage
    public Animal(String food, int lifeExpectancy) {
        this.food = food;
        this.lifeExpectancy = lifeExpectancy;
    }
```

Class Animal

- Getters and Setters

```java
    // Getters and Setters
    public String getFood() {
        return food;
    }

    public void setFood(String food) {
        this.food = food;
    }

    public int getLifeExpectancy() {
        return lifeExpectancy;
    }

    public void setLifeExpectancy(int lifeExpectancy) {
        this.lifeExpectancy = lifeExpectancy;
    }
}
```

Class Animals

- toString();

```java
    // toString method
    1 override
    public String toString() {
        return "This " + food + "-eater has a life expectancy of " + lifeExpectancy + " years.";
    }
```

Class Lion, sub-class (Lion extends Animal)

- Variable

```
class Lion extends Animal {
    10 usages
    private static int age;
    10 usages
    private static String name;
```

Class Lion, sub-class (Lion extends Animal)

- Constructors

```java
// Constructor for age
1 usage
public Lion(int age) {
    super();
    this.age = age;
    this.name = "unknown";
}

// Constructor for name
1 usage
public Lion(String name) {
    super();
    this.age = 0;
    this.name = name;
}

// Constructor for food, life expectancy, age, and name
2 usages
public Lion(String food, int lifeExpectancy, int age, String name) {
    super(food, lifeExpectancy);
    this.age = age;
    this.name = name;
}
```

```java
public Lion() {
    super();
    this.age = 0;
    this.name = "unknown";
}

// Constructor for food, age, and name
1 usage
public Lion(String food, int age, String name) {
    super(food);
    this.age = age;
    this.name = name;
}

// Constructor for life expectancy, age, and name
1 usage
public Lion(int lifeExpectancy, int age, String name) {
    super(lifeExpectancy);
    this.age = age;
    this.name = name;
}

// Constructor for age, and name
public Lion(int age, String name) {
    super();
    this.age = age;
    this.name = name;
}
```

Class Lion, sub-class (Lion extends Animal)

- Getters and Setters

```java
    // Getters and Setters
    public int getAge() {
        return age;
    }

    public void setAge(int age) {
        this.age = age;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }
}
```

Class Main

- Variables

```java
import java.util.Scanner;

public class Main {
    2 usages
    private static String food;
    2 usages
    private static int lifeExpectation;
    2 usages
    private static int age;
    2 usages
    private static String name;
    2 usages
    private static String input;
    5 usages
    private static Scanner scanner = new Scanner(System.in);
```

Class Main

- My Lions

```java
public static void main(String[] args) {

    System.out.println("My Lions:");
    Lion myLion = new Lion( food: "meat", lifeExpectancy: 10, age: 5, name: "Leo");
    System.out.println(myLion.toString());
    Lion myLion1 = new Lion( name: "meat");
    System.out.println(myLion1.toString());
    Lion myLion2 = new Lion( age: 10);
    System.out.println(myLion2.toString());
    Lion myLion3 = new Lion( food: "meat", age: 5, name: "Leo");
    System.out.println(myLion3.toString());
    Lion myLion4 = new Lion( lifeExpectancy: 10, age: 5, name: "Leo");
    System.out.println(myLion4.toString());
```

Class Main

- User’s Lions

```java
        while (true) {
            System.out.println("\nWrite something to begin start //'exit' - for exit");
            input = scanner.next();

            if (input.equalsIgnoreCase( anotherString: "exit")) {
                break;
            }
            System.out.println("Input name: ");
            name = scanner.next();
            System.out.println("Input food: ");
            food = scanner.next();
            System.out.println("Input lifeExpectation: ");
            lifeExpectation = scanner.nextInt();
            System.out.println("Input age: ");
            age = scanner.nextInt();

            Lion lion = new Lion(food, lifeExpectation, age, name);
            System.out.println(lion.toString());

            System.out.println("\n\nDo you want to re-run the program?");
        }
    }
```

Output

- Testing

```
/Users/dany/Documents/GitHub/projects-in-bife/OOP/out/production/OOP skillsDemoTwo.Main
My Lions:
This meat-eater has a life expectancy of 10 years. Its name is Leo and it is 5 years old.
This unknown-eater has a life expectancy of 0 years. Its name is meat and it is 0 years old.
This unknown-eater has a life expectancy of 0 years. Its name is unknown and it is 10 years old.
This meat-eater has a life expectancy of 0 years. Its name is Leo and it is 5 years old.
This unknown-eater has a life expectancy of 10 years. Its name is Leo and it is 5 years old.

Write something to begin start //'exit' - for exit
start
Input name:
Symbo
Input food:
meat
Input lifeExpectation:
150
Input age:
12
This meat-eater has a life expectancy of 150 years. Its name is Symbo and it is 12 years old.


Do you want to re-run the program?

Write something to begin start //'exit' - for exit
exit

Process finished with exit code 0
```

- I used my OOP knowledge to create this app. I used the principles of OOP.